higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

ENGINEERING FABRICATION – SHEET METAL WORK
NQF LEVEL 4

(6031024)

12 March 2019 (Y-Paper)
13:00–16:00

This question paper consists of 6 pages.

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TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Write neatly and legibly.

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QUESTION 1

1.1 State TWO advantages of using rigid fibrous duct board when manufacturing long-lasting, light ducts. (2)

1.2 Describe shortly the term *seam* as used in fabrication. (2)

1.3 State FOUR criteria that must be followed as a general principle when air diffusion terminals within ductwork-layout are selected and located. (4)

1.4 Explain what is meant by the term *acoustic acquisition for ductwork*. (2)

1.5 When designing the layout of a plant, always make sure that the layout has enough space around components or equipment to carry out normal service and maintenance.

State FIVE examples of components or equipment that applies to the above statement. (5)

[15]

QUESTION 2

2.1 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (2.1.1–2.1.3) in the ANSWER BOOK.

2.1.1 When drawings are laid out for ductwork, the correct symbols must be used.

2.1.2 Galvanised sheet metal is commonly used in ductwork because it does not rust.

2.1.3 Shiplap is classified as a resin.

(3 × 1) (3)

2.2 Line work in duct drawings is very important. Unclear lines result in poor quality drawings, especially when copied.

State the FOUR important qualities of clear lines. (4)

2.3 A grill is an important part in ductwork.

Define the term *grill*. (3)

2.4 Name FIVE types of components or fittings that will be needed for ductwork. (5)

[15]

QUESTION 3

3.1 TIG *welding* is also known as GTAW.

Give the meaning of any TWO of the letters in the abbreviation GTAW.

(2 × 1) (2)

3.2 FIGURE 1 below indicates TIG welding in process.

Name each of the labelled parts in FIGURE 1. Write only the answer next to the letter (A–E) in the ANSWER BOOK.

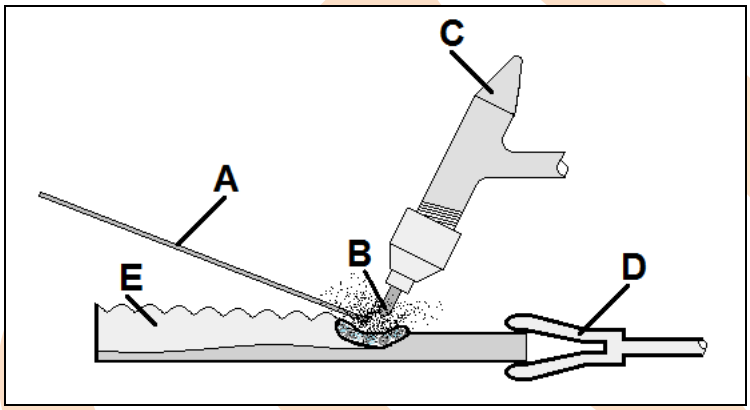


FIGURE 1 (5)

3.3 The type of gas used during TIG welding is of vital importance.

Briefly explain the purpose of a shielding gas used during gas welding. (3)

3.4 Can oxygen be used as a shielding gas for TIG welding? (1)

3.5 The table below shows welding defects and their preventative measures.

Choose a preventative measure from COLUMN B that matches a defect in COLUMN A. Write only the letter (A–D) next to the question number (3.5.1–3.5.4) in the ANSWER BOOK.

COLUMN A		COLUMN B	
3.5.1	Poor fusion	A	use different gas
3.5.2	Spatter	B	use higher welding current
3.5.3	Porosity	C	decrease current
3.5.4	Undercut	D	keep puddle melted for a longer period of time

(4 × 1) (4)

[15]

QUESTION 4

4.1 Creating the arc during TIG welding is a difficult process, specifically for the beginner welder.

List the THREE procedures to be followed when a welder strikes an arc during TIG welding in sequence. (3)

4.2 Name the THREE elements on which the flow rate of shielding gas in TIG welding depends on. (3)

4.3 State the SEVEN main points that safety precautions focus on in the TIG welding workshop. (7)

4.4 The earth (ground) clamp is a cathode of the circuit when welding aluminium.

Explain the word *cathode*. (2)

[15]

QUESTION 5

5.1 State EIGHT functions of a post processor. (8)

5.2 By means of drawings, indicate the following important machine positions/points that are needed to program a CNC machine:

5.2.1 Reference point 'R'

5.2.2 Tool reference point 'N'

5.2.3 Zero point of the machine 'M'

(3 × 2) (6)

5.3 Explain the following terms related to CNC machining:

5.3.1 Software

5.3.2 Hardware

5.3.3 Geometric algorithm

5.3.4 Toolpath

(4 × 2) (8)

5.4 Conventional lathe machines and CNC lathe machines have some features in common.

List FOUR common features that are found in both CNC and conventional lathe machines. (4)

5.5 Explain the *editing procedure* with reference to a CNC program. (4 × 1) (4)

5.6 The table below explains the steps to be followed when operating a CNC machine. Choose TWO descriptions from COLUMN B that matches each relevant step in COLUMN A. Write only the letters (e.g. A & B) next to the question number (5.6.1–5.6.5) in the ANSWER BOOK.

COLUMN A		COLUMN B	
5.6.1	Step 1: Select the correct tools for the specific job.	A	removing and measuring machine components is standard machine shop practice
5.6.2	Step 2: Check, adjust and replace worn or damaged tooling.	B	use normal workshop cleaning materials
5.6.3	Step 3: Troubleshoot common machine faults during and after operation.	C	the amount of material that must be removed
5.6.4	Step 4: Remove and measure machine components.	D	removing components is usually a straightforward task and depends on the clamping methods used
5.6.5	Step 5: Clean the machine.	E	inappropriate coolant and coolant flow
		F	if the tool is slightly worn, this will cause incorrect dimensions
		G	misalignment of the tool holder in the CNC machine spindle
		H	ensure that the machine is completely switched off
		I	the cutting speed you require
		J	if the tool is blunt, the material will burn

(5 × 2) (10)
[40]

TOTAL: 100

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